

Validity of the price-taking assumption, the core

Price-taking assumption

At least two approaches

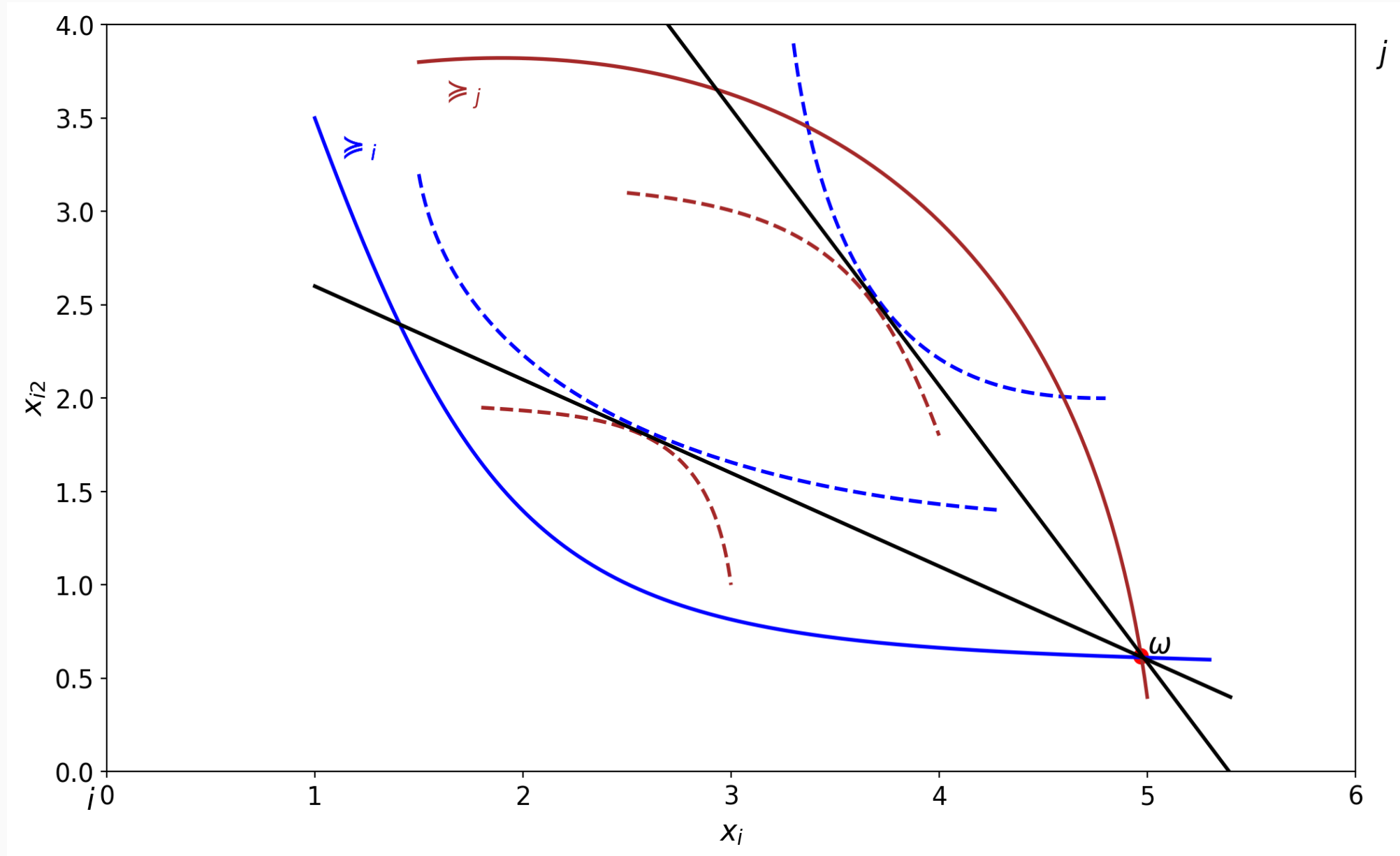
- Theory of price formation. Non-cooperative approach.
- Postulate outcome of markets in terms of allocations and compare to WE.

Edgeworth (1881) argued that the main features of markets are

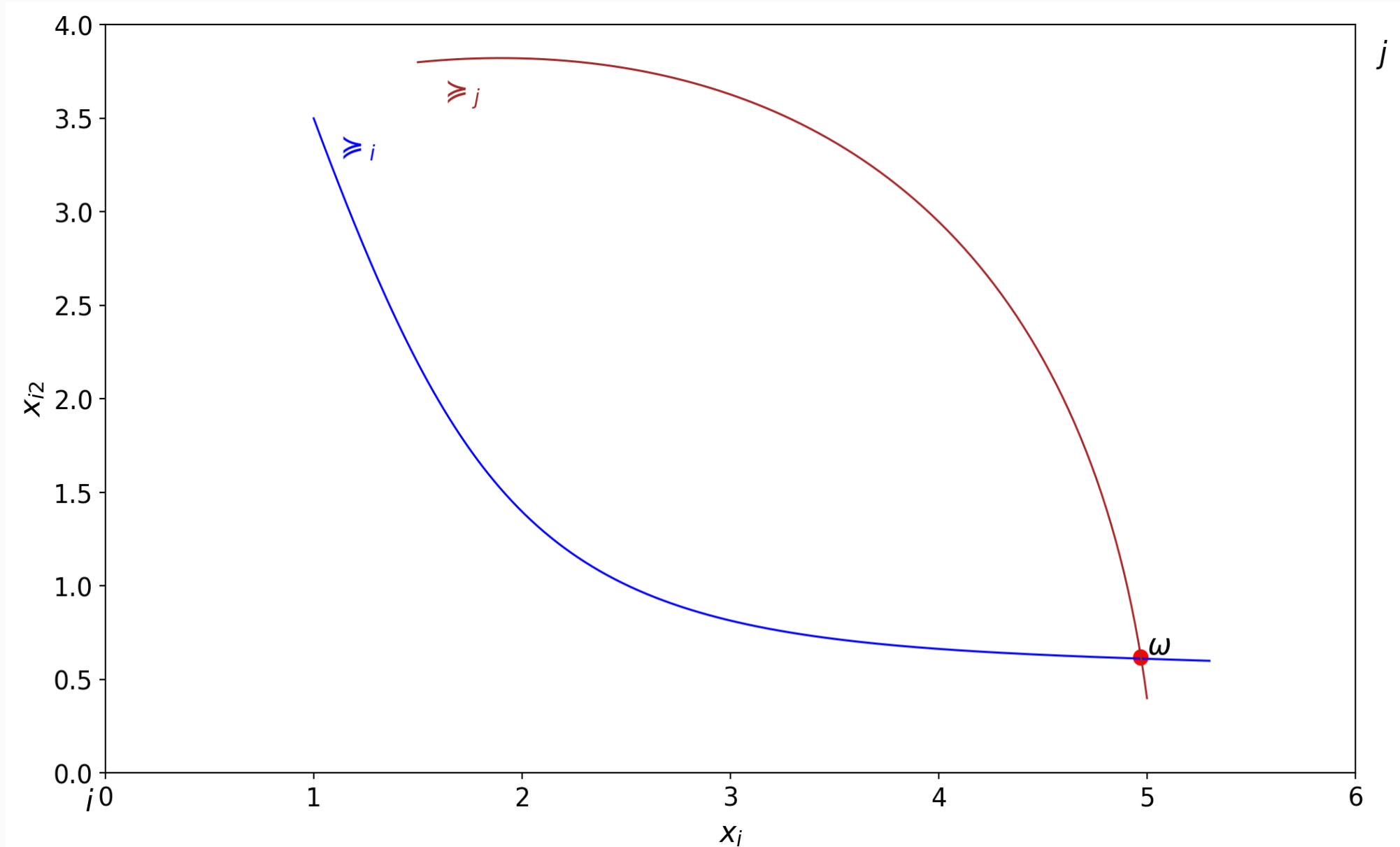
1. Trade is voluntary, and
2. All gains from trade are exhausted.

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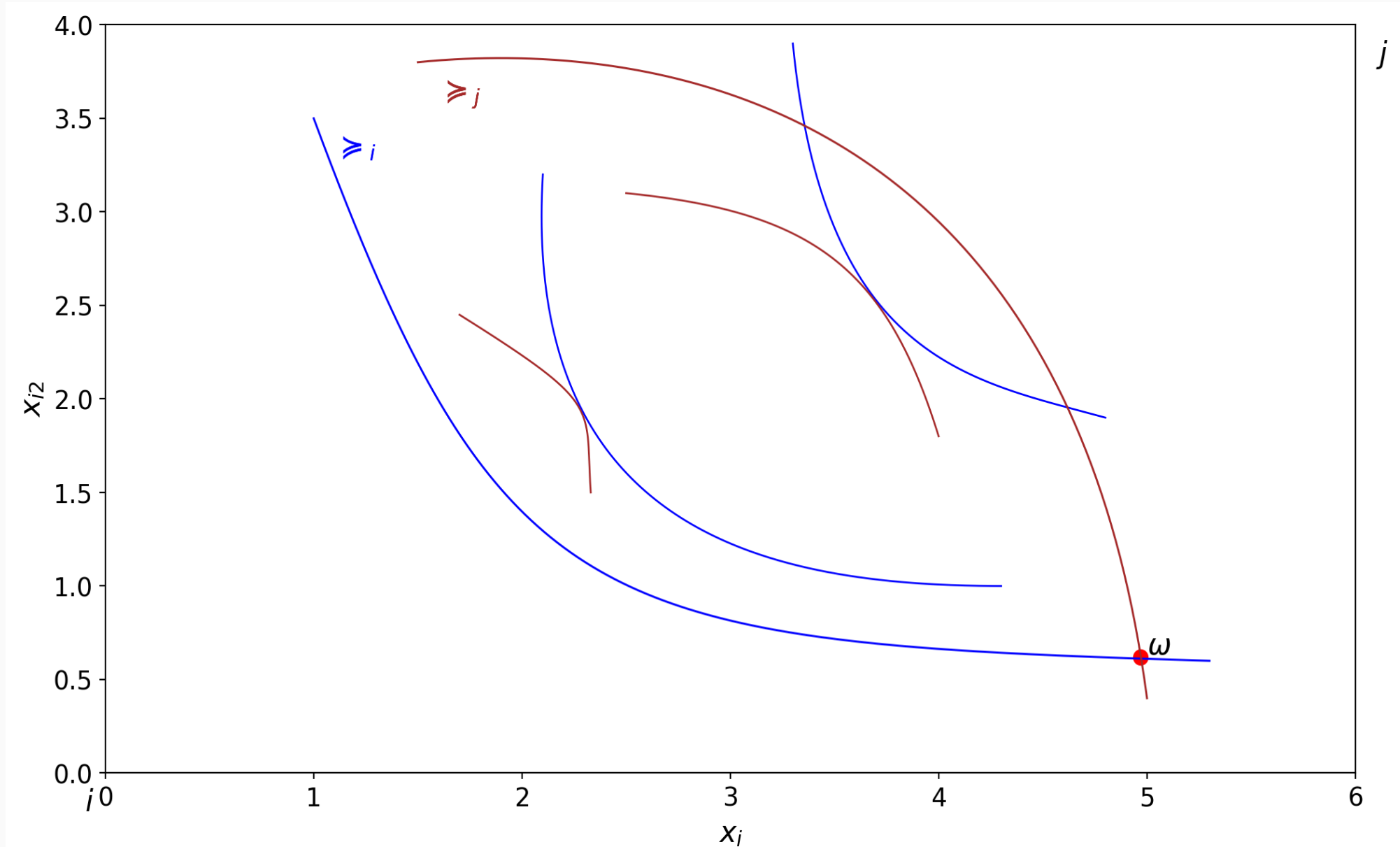
Two Walrasian equilibria



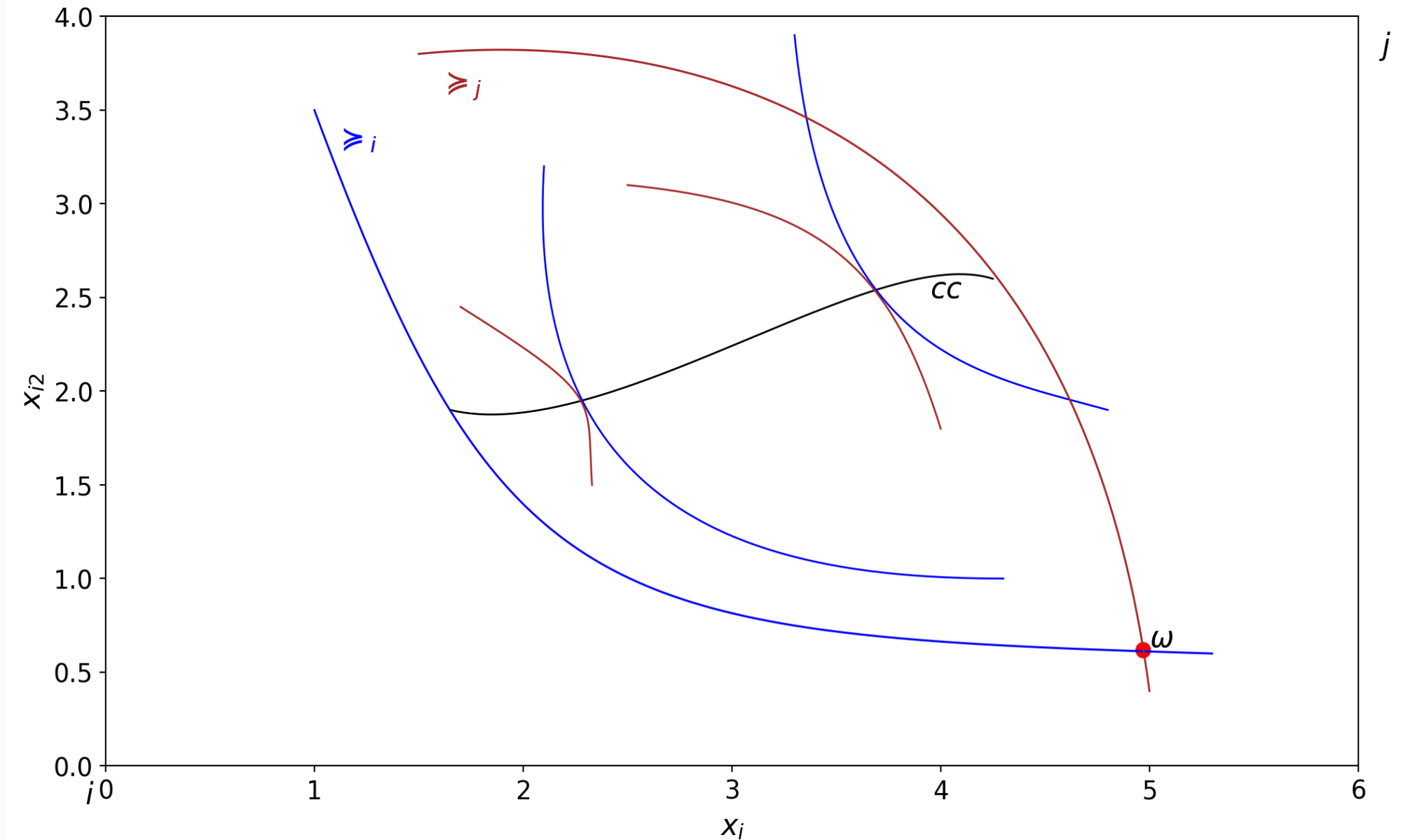
Edgeworth conjecture



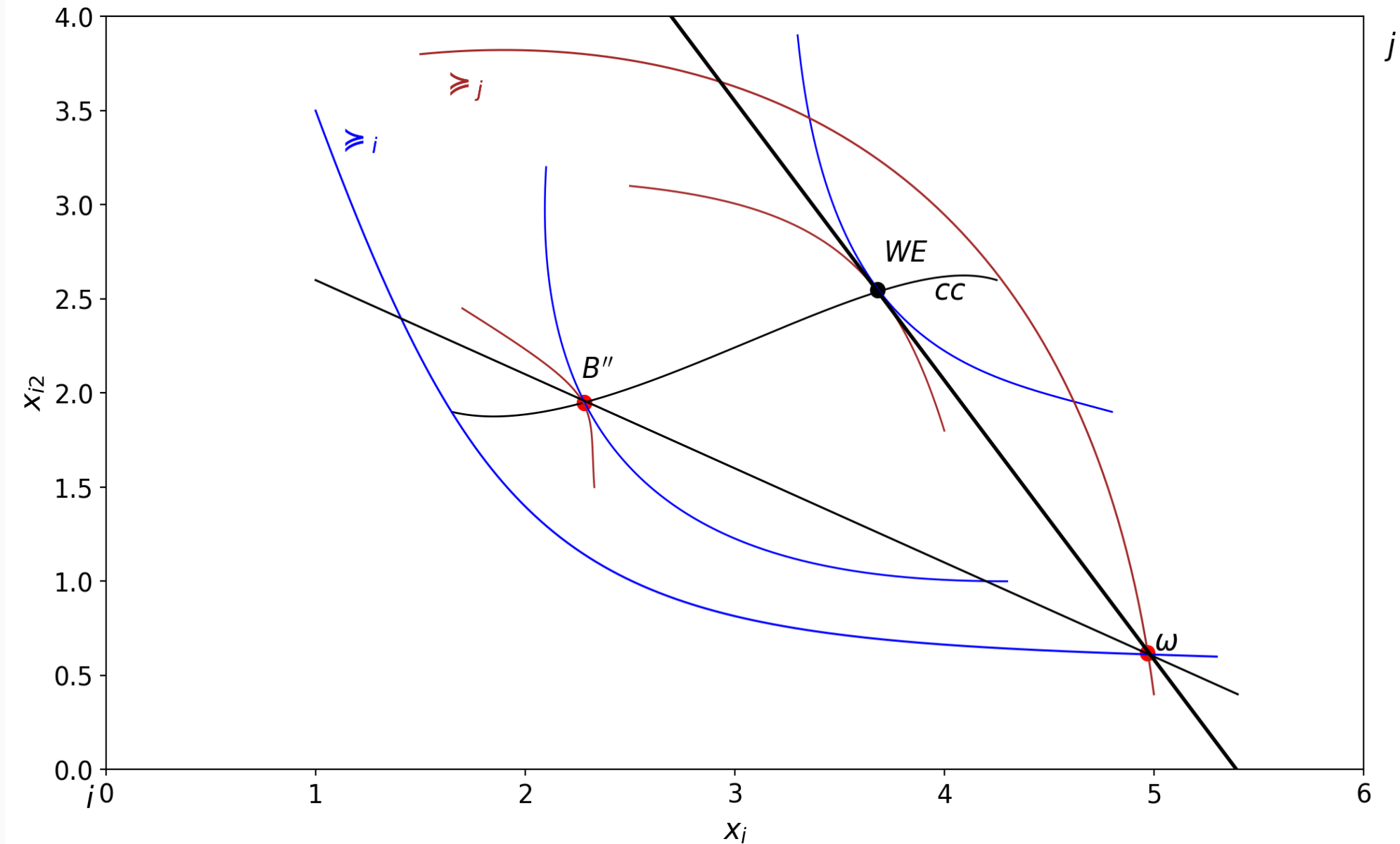
Edgeworth conjecture 2



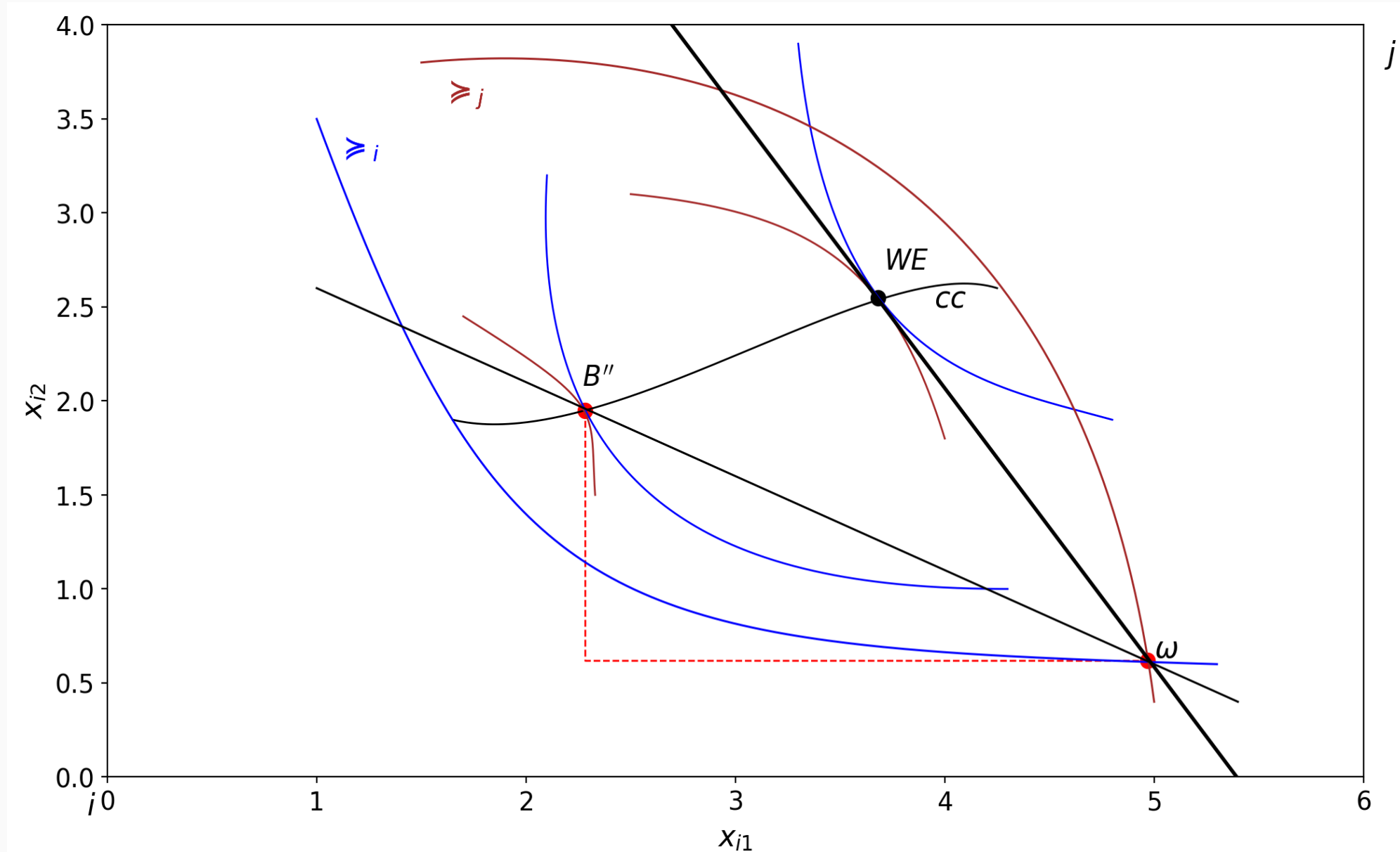
Edgeworth conjecture 3



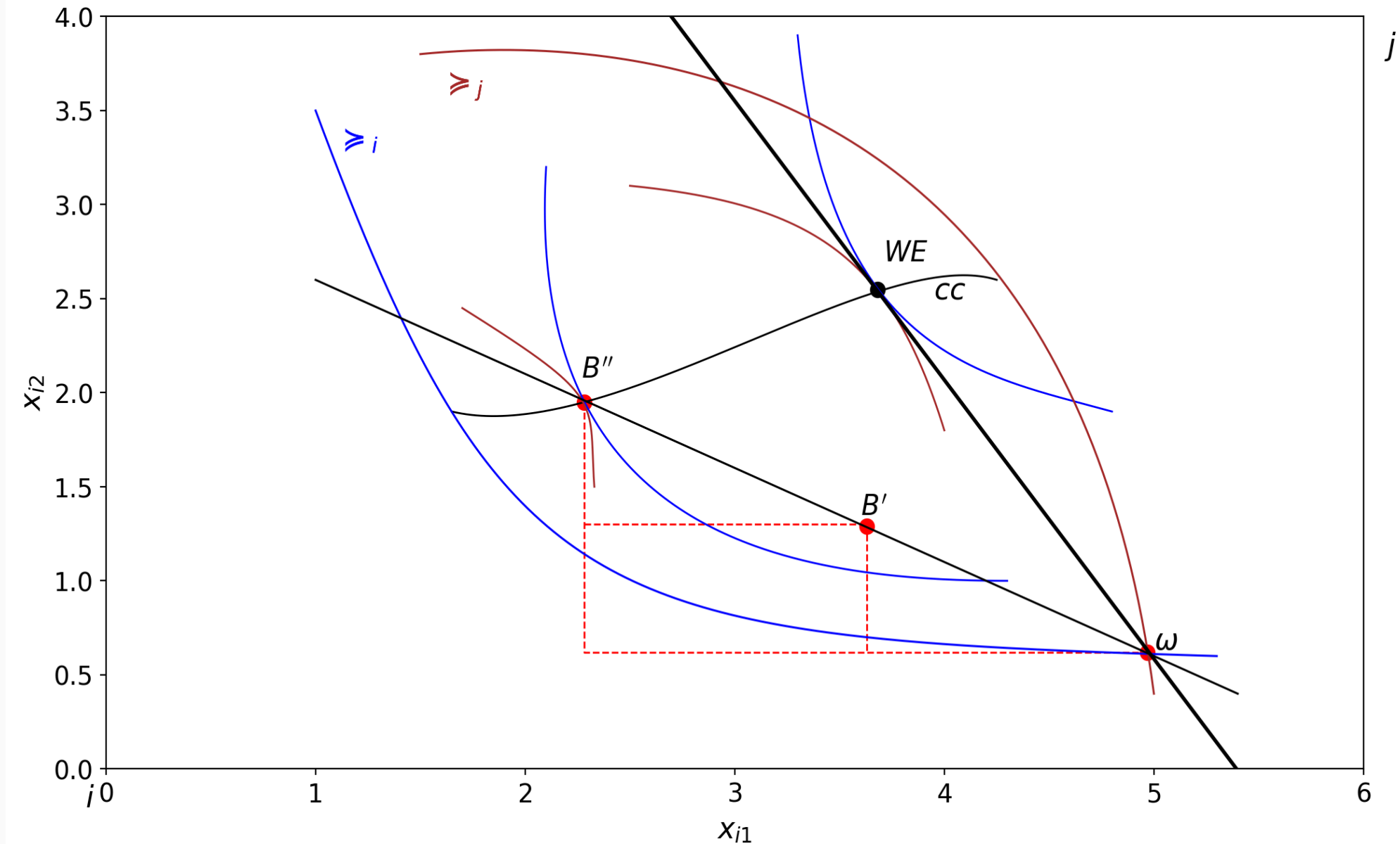
Edgeworth conjecture 4



Edgeworth conjecture 5



Edgeworth conjecture 6



Two agents of each type

- The economy has now 4 agents: i_1, i_2 and j_1, j_2 are “replicas” of i and j respectively.
- B'' is not a “market outcome” because it does not exhaust all gains from trade.
- If i_1, i_2 and j_1 trade, i_1, i_2 may obtain B' while j_1 obtains B'' .
- $2 \times (\omega_i - B') = \omega_j - B''$. Feasibility.
- j_1 consumes B'' and is as well off as before.
- i_1 and i_2 consume B' are better off than at B'' .

The core of an economy

Fix a pure exchange economy $\mathcal{E} = (L, \{X_i, \succeq_i, \omega_i\}_{i=1}^I)$.

Definition 1 Let $\mathcal{J} = \{1, 2, \dots, I\}$. Every $A \subseteq \mathcal{J}$ is a coalition. $\mathcal{J}$ is known as the grand coalition.

Definition 2 $A \subseteq \mathcal{J}$ blocks an allocation x if $(\forall i \in A) (\exists x'_i \in X_i)$

1. $\sum_{i \in A} x'_i = \sum_{i \in A} \omega_i$,
2. $x'_i \succeq_i x_i$, and
3. $(\exists i' \in A) x'_{i'} \succ_{i'} x_{i'}$.

Definition 3 (Core) The core of $\mathcal{E}$, denoted $\mathcal{C}(\mathcal{E})$, is the set of allocations that cannot be blocked by any coalition.

Notation.

- Occasionally, $f : \mathcal{I} \rightarrow \mathbb{R}^L$ is used to denote a consumption assignment.

For instance, if $f \in \mathcal{C}(\mathcal{E})$, f maps $i \mapsto f(i) \in X_i$. Thus the bundle x_i assigned to agent i is denoted $f(i) = x_i$ or $x(i)$.

- When the economy studied is clear, $\mathcal{C}$ may be used instead of $\mathcal{C}(\mathcal{E})$, to denote its core.

The core

Theorem 1 If an allocation is in the core then it is Pareto optimal.

Proof. Let $x \in \mathcal{C}(\mathcal{E})$. $\mathcal{J}$ does not block x .

This means x is P.O.

Theorem 2 If preferences are monotone, then every WE allocation is in the core.

Proof.

- Let $(\{x_i\}_{i \in \mathcal{I}}, p)$ be a Walrasian equilibrium.
- Suppose $x = \{x_i\}_{i \in \mathcal{I}} \notin \mathcal{C}(\mathcal{E})$. Then $(\exists B \subseteq \mathcal{I})$:
 - $\exists \{y_i\}_{i \in B} : \sum_{i \in B} y_i = \sum_{i \in B} \omega_i$,
 - $(\forall i \in B) y_i \succeq_i x_i$ and
 - $(\exists j \in B) y_j \succ_j x_j$
 - $(\forall i \in B) p \cdot y_i \geq p \cdot x_i$ and $p \cdot y_j > p \cdot x_j$

QED.

$$\begin{aligned}
\sum_{i \in B} p \cdot y_i &> \sum_{i \in B} p \cdot x_i \\
p \cdot \sum_{i \in B} y_i &> \sum_{i \in B} p \cdot x_i \\
&= \sum_{i \in B} p \cdot \omega_i \quad (\text{by } x_i \in D_i(p, \omega_i) \wedge p \cdot x_i = p \cdot \omega_i) \\
&= p \cdot \sum_{i \in B} \omega_i \\
&= p \cdot \sum_{i \in B} y_i
\end{aligned}$$

The 3rd line uses monotonicity. The contradiction completes the proof.

Replica sequences of economies

- Let T be a finite set of consumers, that is of pairs $(\succeq_t, \omega_t)$. T is a pure exchange economy.
- Let T_n be the n^{th} replica of T , that is to say, the economy where there are n agents of each type $(\succeq_t, \omega_t) \in T$.

Debreu and Scarf

- Debreu and Scarf (1963)
- Suppose that preferences are strongly convex, locally non-satiated, complete, and transitive.
- Equal Treatment Theorem (Debreu and Scarf 1963). In any core allocation of T_n , agents of the same type must receive the same core allocation.
- Theorem (Debreu and Scarf 1963). If an allocation belongs to the core of T_n for all n , then it is a Walrasian quasi-equilibrium.

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- The modern definition of the core is due to Gillies (1959).
 - Debreu and Scarf (1963) first formalize and prove a general form of Edgeworth's conjecture for replica sequences of economies.
 - Aumann (1964) formulates the problem with a continuum of agents and shows that the core coincides with the set of Walrasian equilibria.
Aumann assumed monotonicity but it is immediate from his proof, that LNS suffices to show that core allocations are Walrasian quasi-equilibria.

Core convergence

- The following is based on Anderson (1978).
- Given a finite set $\mathcal{J}$, $|\mathcal{J}|$ is the number of elements in $\mathcal{J}$.

Definition 4 Let $\mathcal{P}$ the the set of binary relations $\succ$ on $\mathbb{R}_+^L$ that satisfy

1. Weak monotonicity: $x \gg y \implies x \succ y$
2. Free disposal: $x \gg y$ and $y \succ z \implies x \succ z$

Core convergence

Theorem 3 (Anderson) Let $\{\mathbb{R}_+^L, \succsim_i, \omega_i\}_{i \in \mathcal{I}}$ be a pure exchange economy with finitely many agents and $(\forall i \in \mathcal{I}) \succsim_i \in \mathcal{P}$.

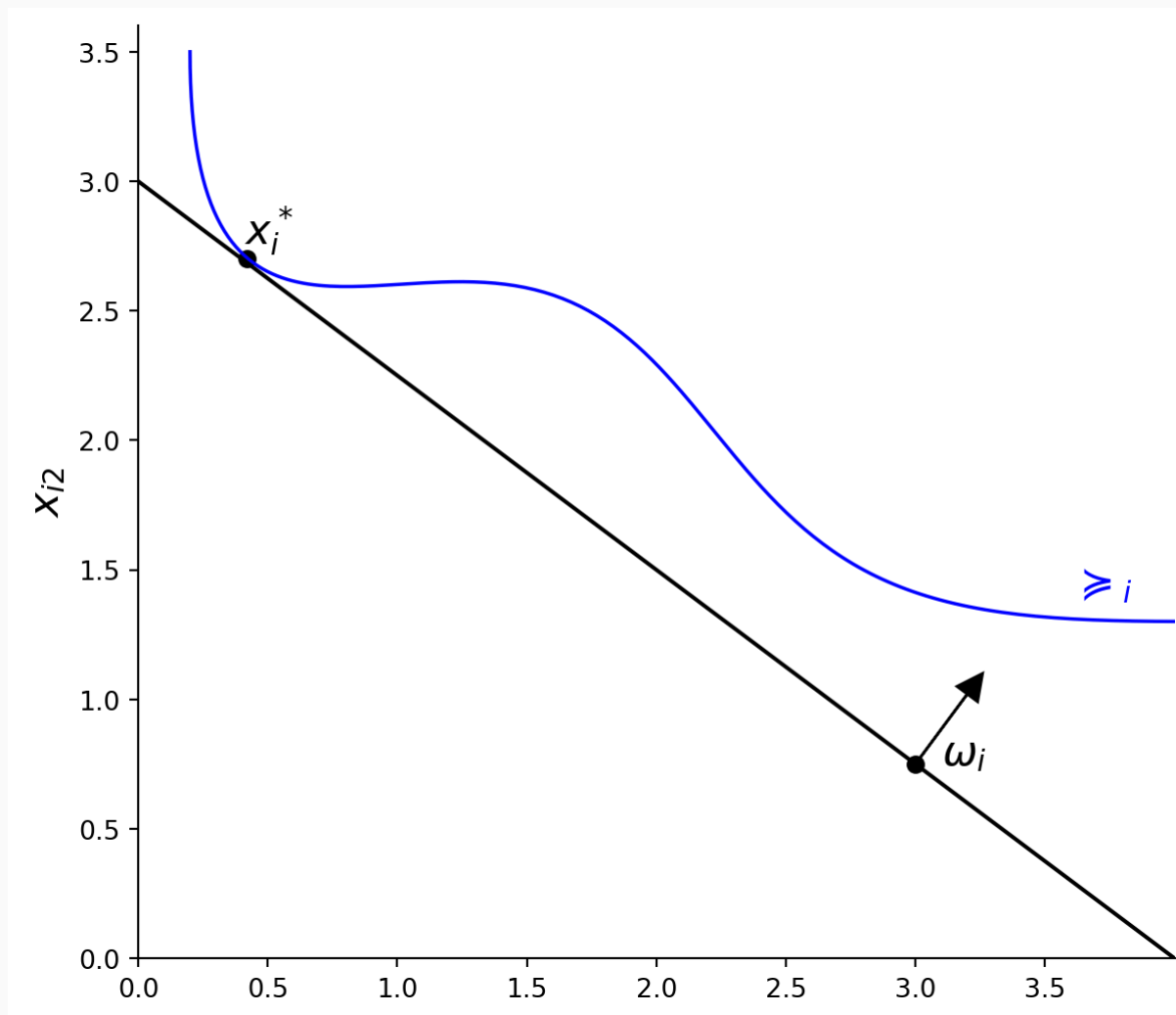
Let

$$M = \sup \left\{ \left\| \sum_{\mathcal{H}} \omega_j \right\|_{\infty} : \mathcal{H} \subseteq \mathcal{I}, |\mathcal{H}| = L \right\}.$$

If $f \in \mathcal{C}$, then $\exists p : \mathbb{R}^L_+ \rightarrow \mathbb{R}$

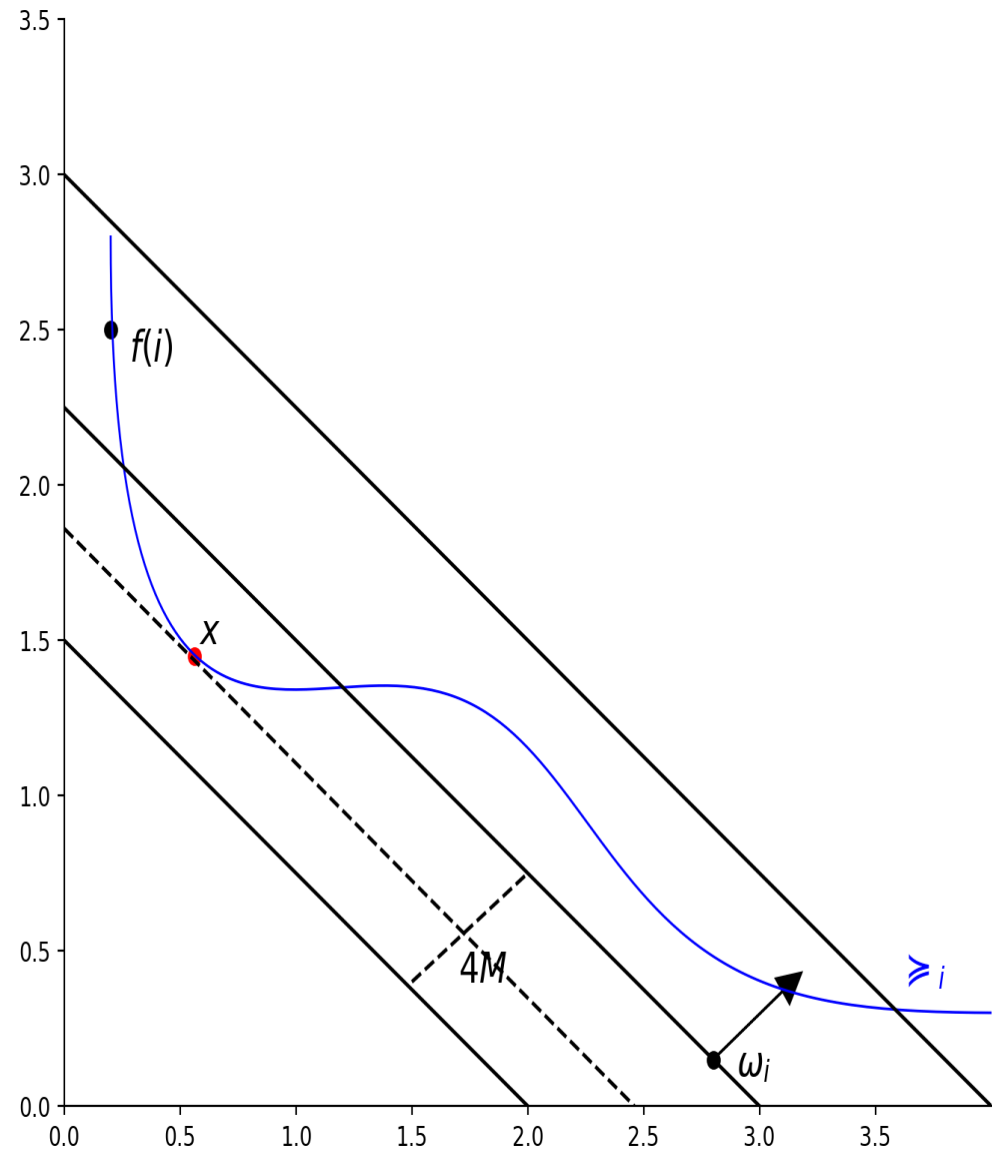
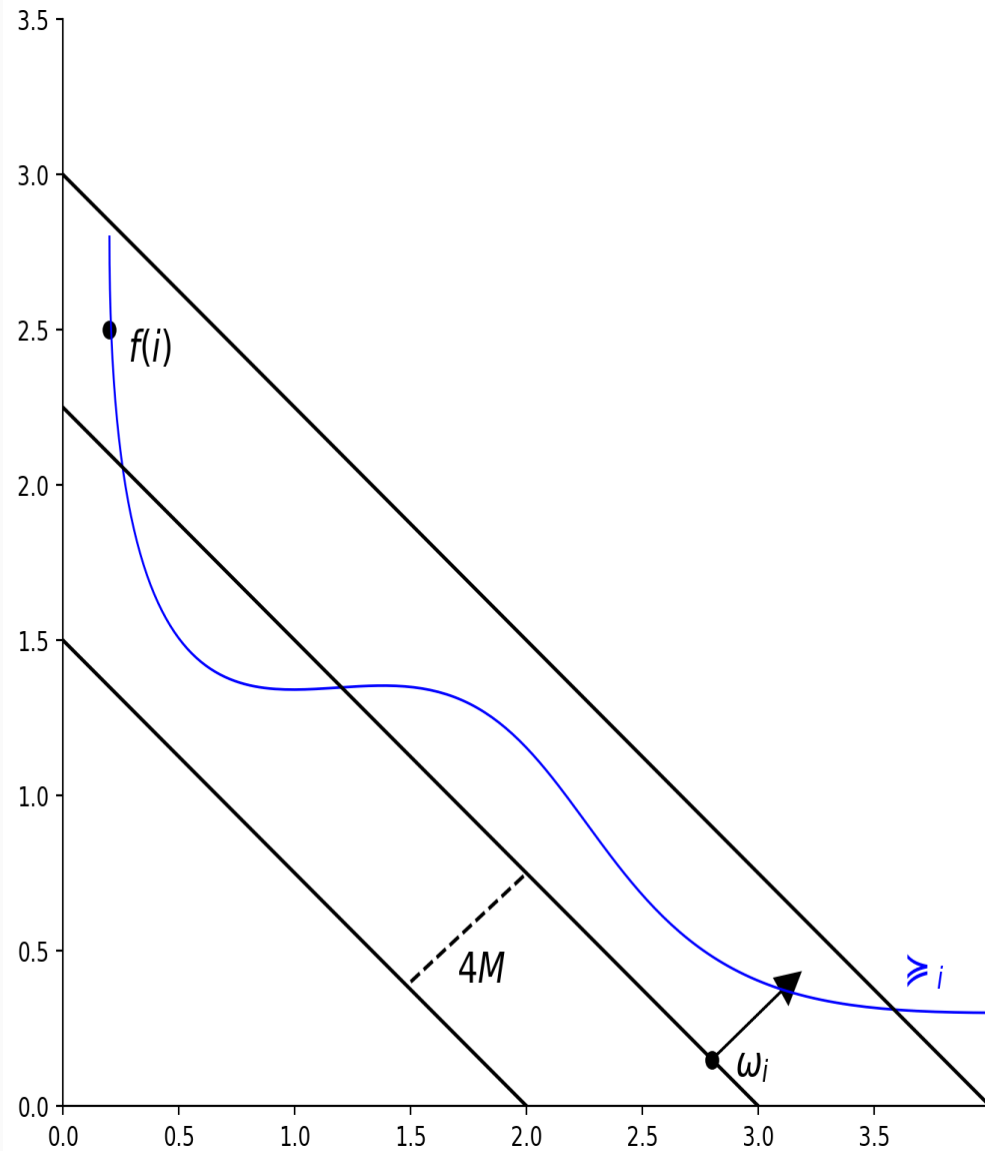
1. $\frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} |p \cdot (f(i) - \omega_i)| \leq \frac{4M}{|\mathcal{I}|}$
2. $\frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} |\inf\{p \cdot (x - \omega_i) : x \succsim_i f(i)\}| \leq \frac{4M}{|\mathcal{I}|}$

Demand



- x_i^* must be affordable; any $x \succ_i x_i^*$ must be too expensive.

Almost demand



Shapley-Folkman theorem, preliminaries

- Let A_1, A_2 be subsets of a vector space X . Then,
- $A_1 + A_2 = \{x_1 + x_2 : x_1 \in A_1, x_2 \in A_2\}$.
- $\text{co } A_1$ is the convex hull of A_1
- If A_1, A_2 are convex, then $A_1 + A_2$ is convex.
- For $x \in \mathbb{R}^L$, $\|x\|_\infty = \max\{|x_\ell| : 1 \leq \ell \leq L\}$.

Theorem 4 (Shapley-Folkman) Let $\mathcal{J} = \{1, 2, \dots, I\}$, $(\forall i \in \mathcal{J}) A_i \subseteq \mathbb{R}^L$, $A = \sum_{i \in \mathcal{J}} A_i$.

Then $(\forall x \in \text{co } A) (\exists B \subseteq \mathcal{J})$ such that

1. $|B| \leq L$

2. $x = a_1 + \dots + a_I$

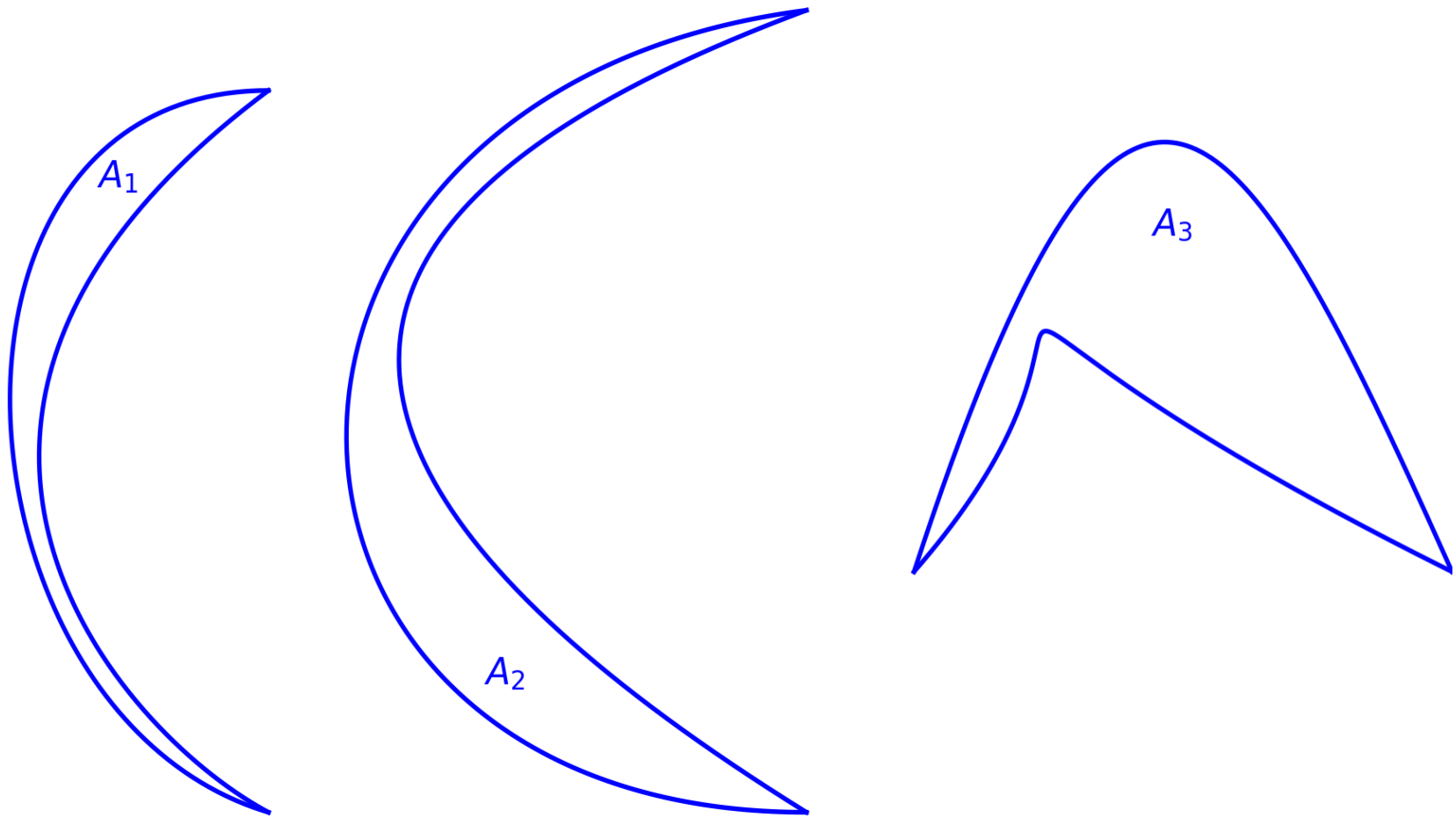
where $(\forall i \in \mathcal{J} \setminus B) a_i \in A_i$ and $(\forall i \in B) a_i \in \text{co } A_i$.

- A version of the theorem appears in Starr (1969). Shapley and Folkman provided its proof in a private communication.

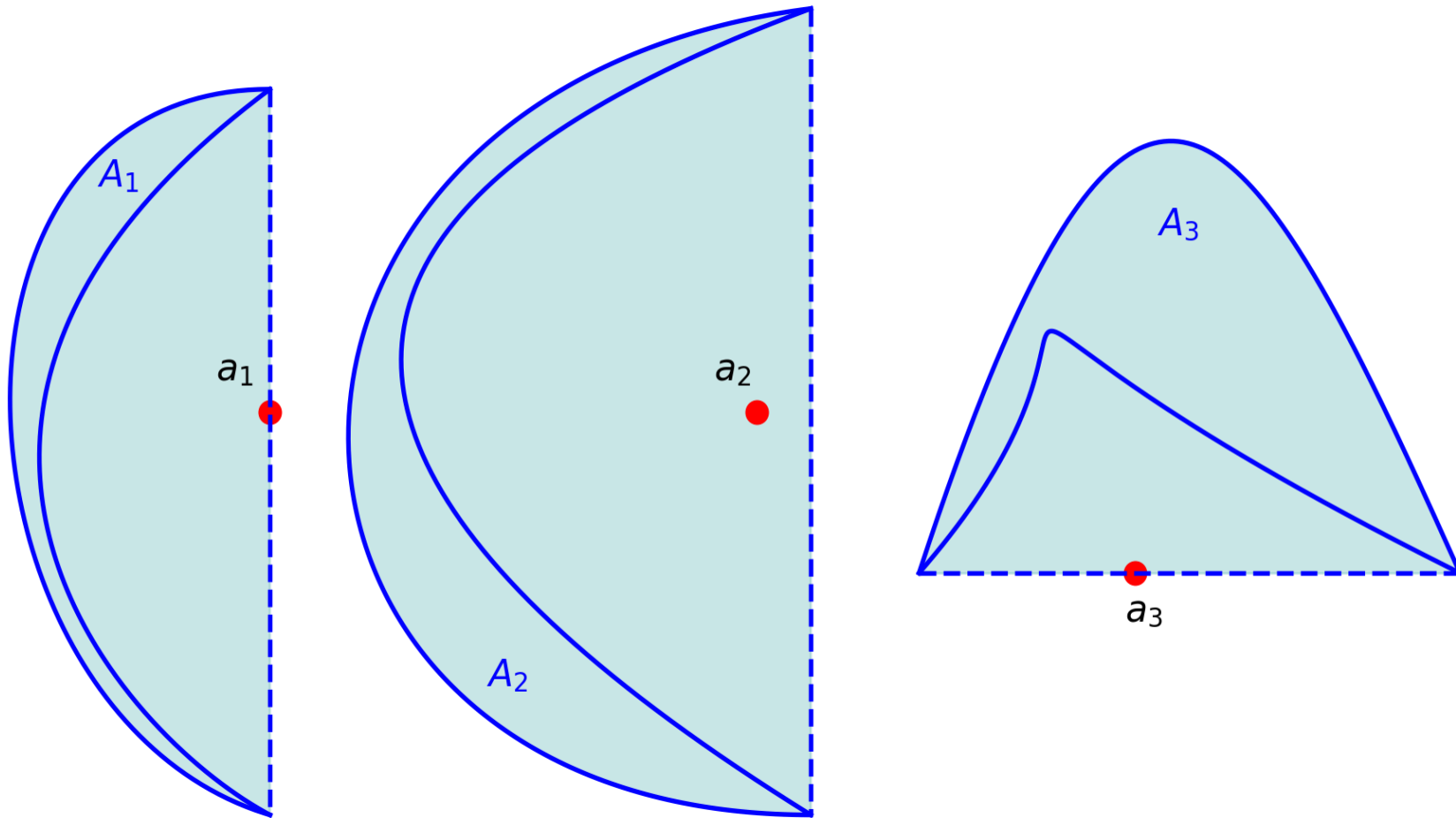
Example

- For $i = 1, 2, 3$, $A_i \subset \mathbb{R}^2$, A_i is not convex.
- Note that $\text{co}(A_1 + A_2 + A_3) = \text{co } A_1 + \text{co } A_2 + \text{co } A_3$.
- Pick $x \in \text{co}(A_1 + A_2 + A_3)$.
- Then $x = a_1 + a_2 + a_3$, $a_i \in \text{co } A_i$.
- There are b_1, b_2, b_3 such that
 - $b_1 \in A_1$,
 - $b_2 \in \text{co } A_2$,
 - $b_3 \in \text{co } A_3$ and
 - $x = b_1 + b_2 + b_3$.

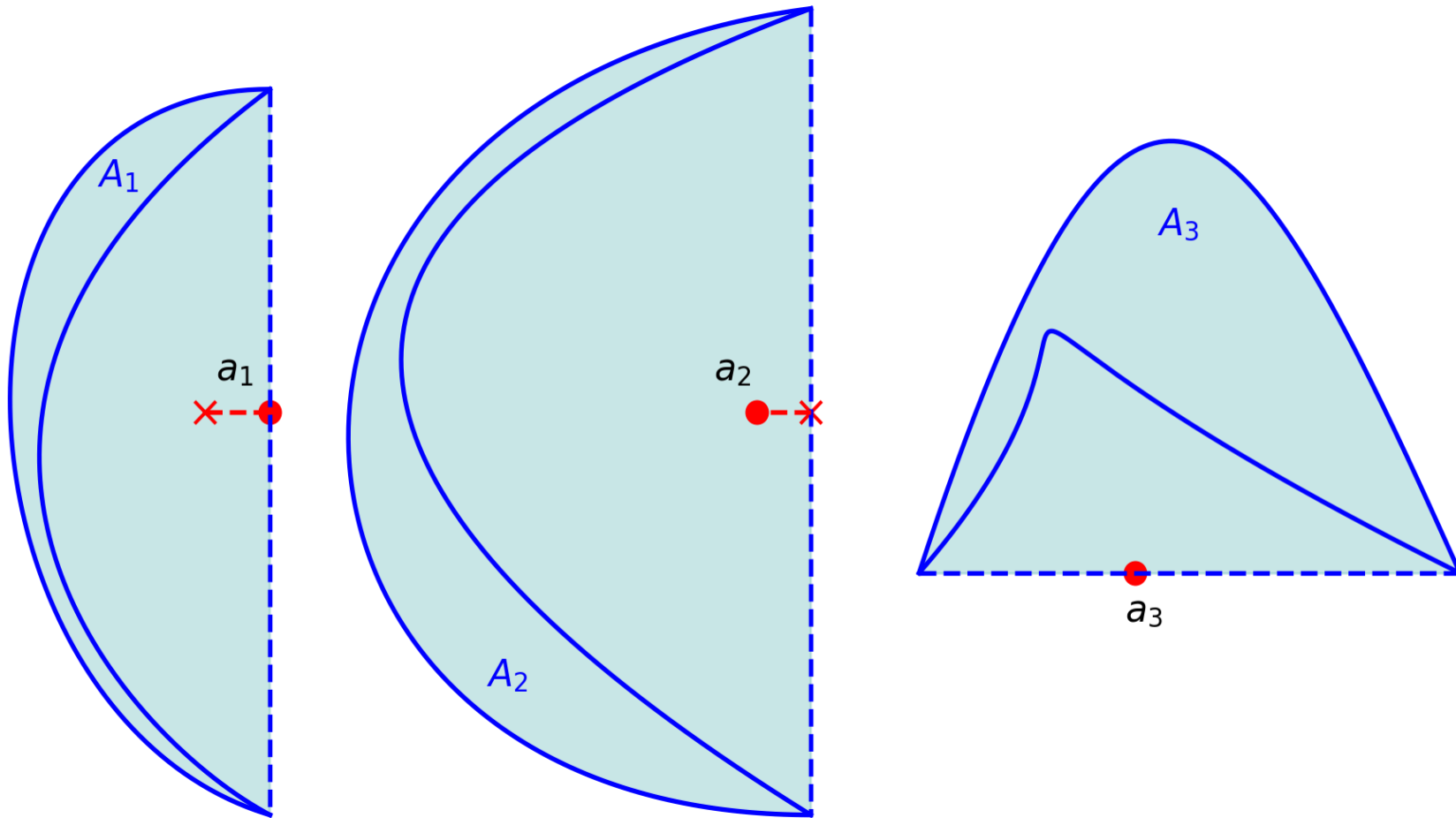
Picture 1



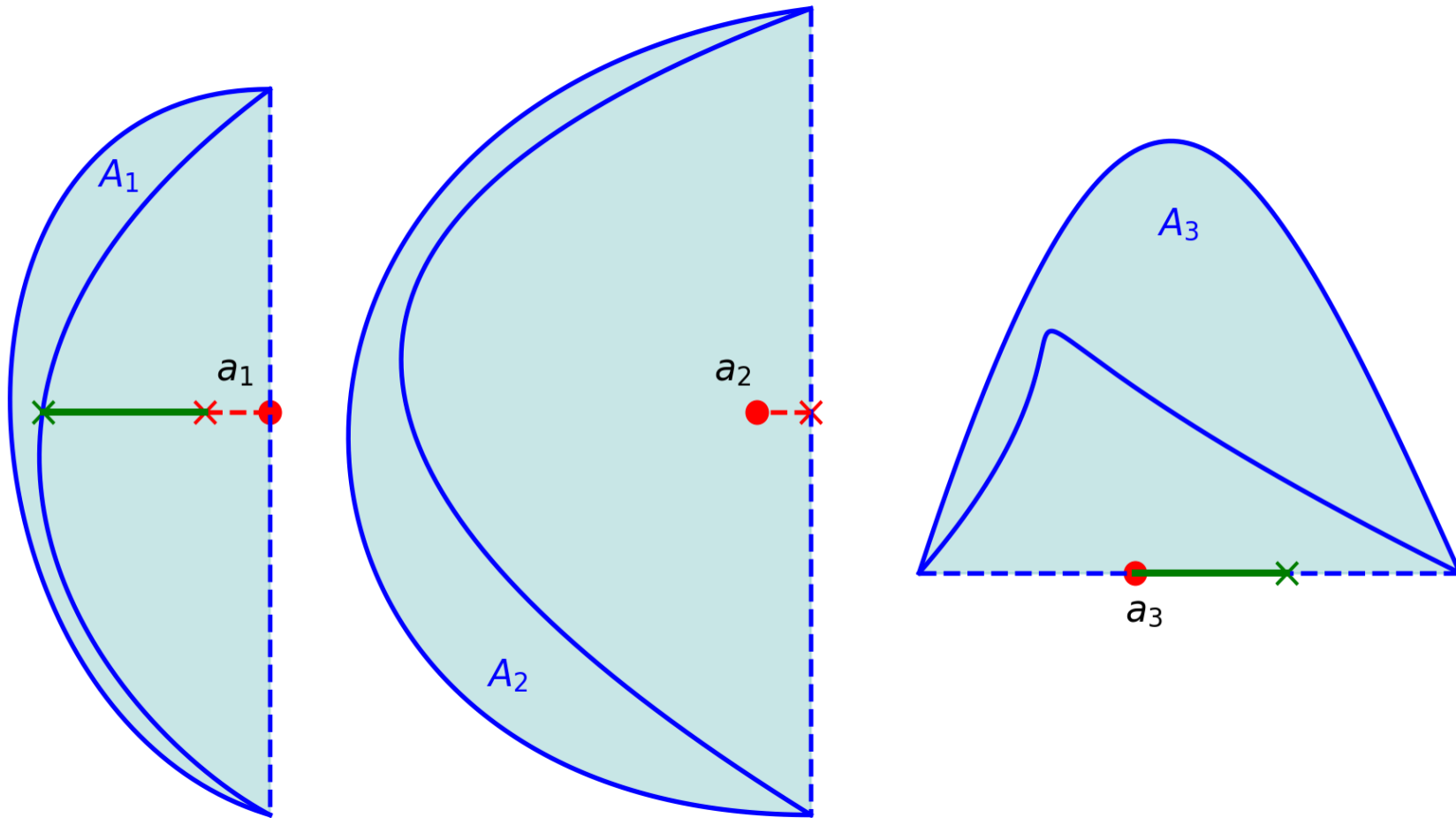
Picture 2



Picture 3



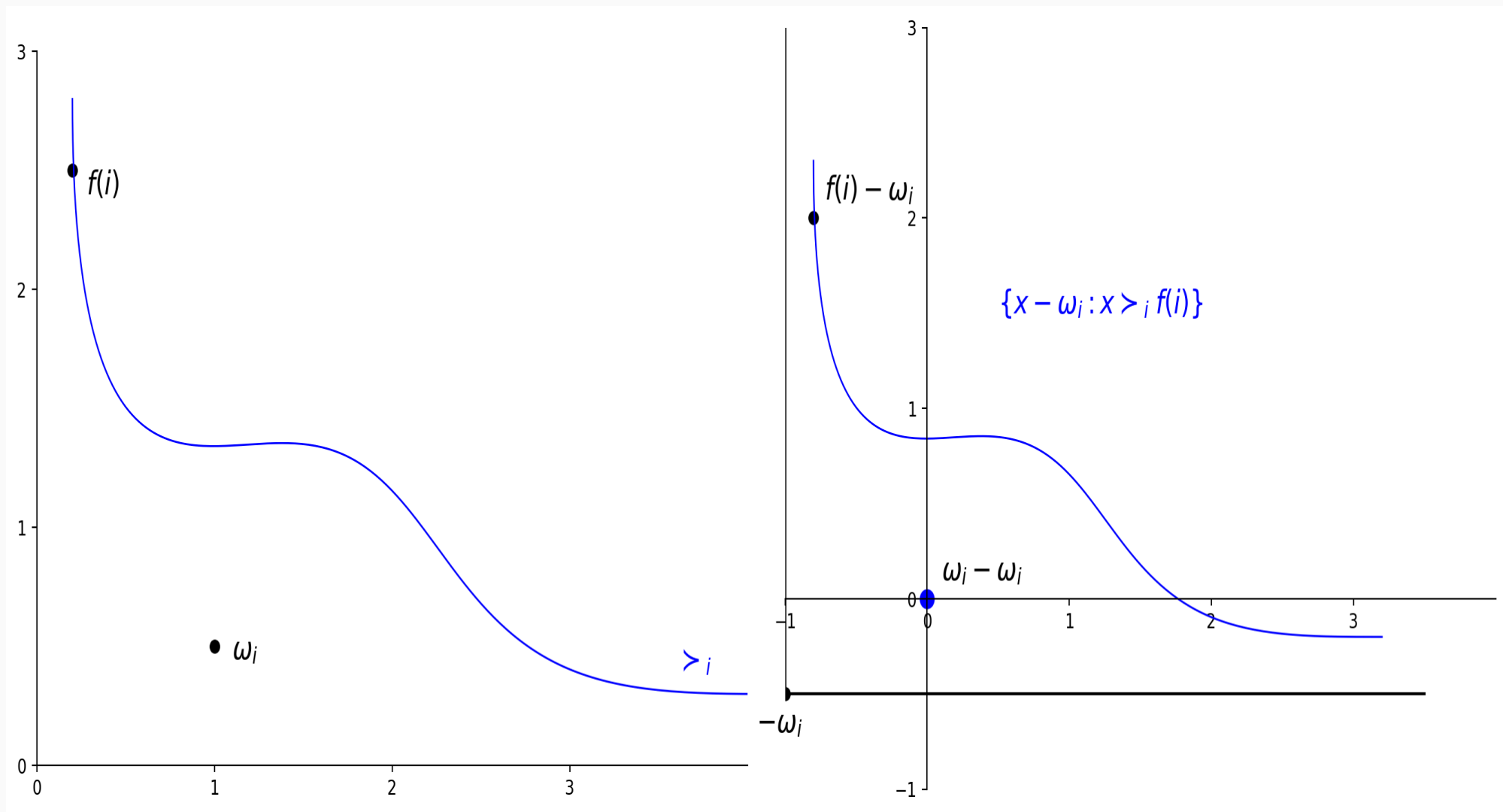
Picture 4



Proof of core theorem

Define

- $\forall i \in \mathcal{I}, \varphi(i) = \{x - \omega_i : x \succ_i f(i)\} \cup \{0\}$
- $\Phi = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \varphi(i)$



Claim. $G \ll 0 \implies G \notin \Phi$.

- Suppose not. Then $\exists g : \mathcal{J} \rightarrow \mathbb{R}^L$ such that

$$(\forall i \in \mathcal{J}) \left[g(i) \in \varphi(i) \wedge G = \frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J}} g(i) \ll 0 \right]$$

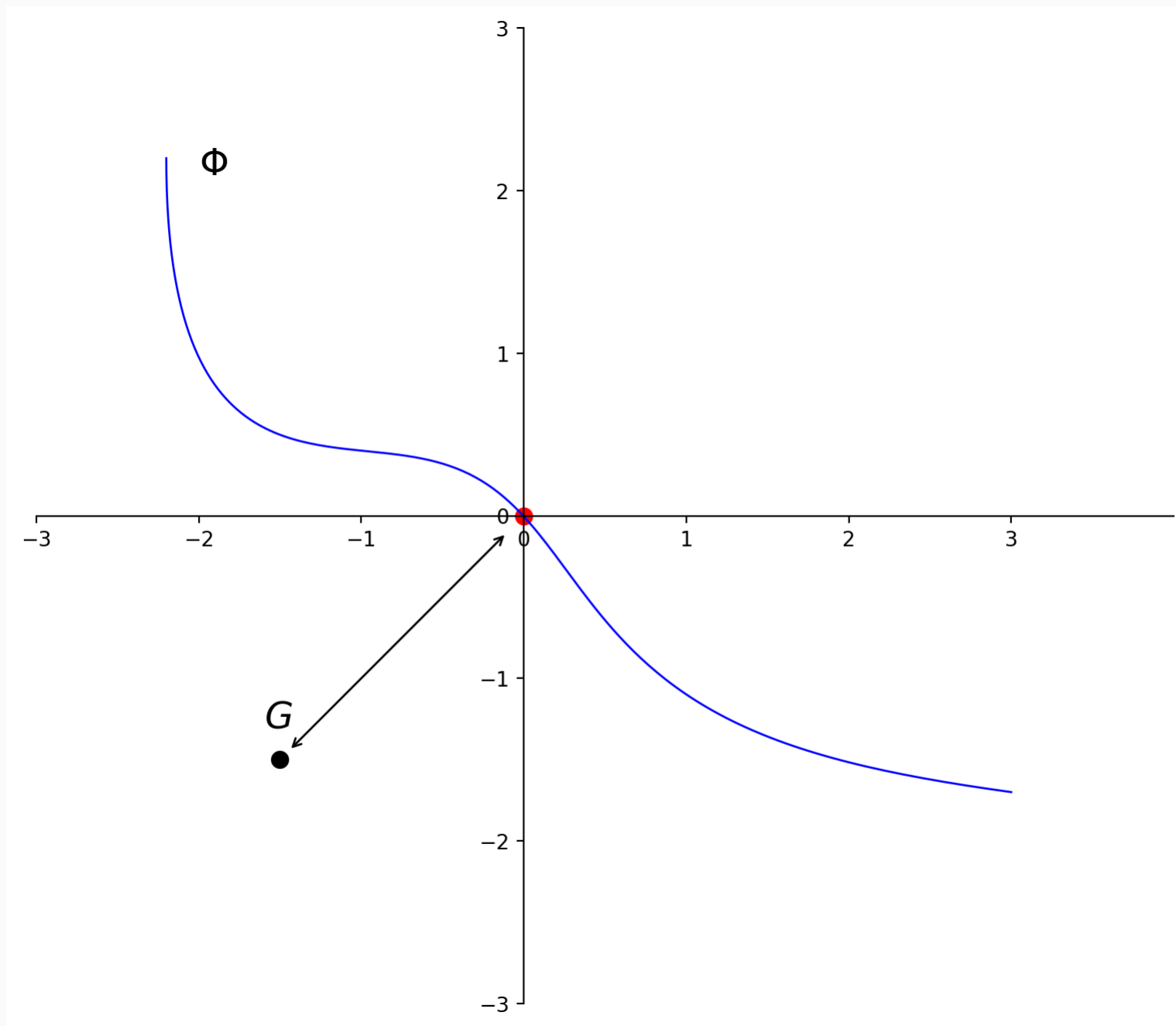
- Let $B = \{i \in \mathcal{J} : g(i) \neq 0\}$ and define

$$(\forall i \in B) \ h(i) = g(i) + \omega_i - G \frac{|\mathcal{J}|}{|B|}$$

- $(\forall i \in B) \setminus [h(i) \gg g(i) + \omega_i \succsim_i f(i)] \implies h(i) \succsim_i f(i)$

$$\begin{aligned}\sum_B h(i) &= \sum_B g(i) + \sum_B \omega_i - |B| \frac{|\mathcal{J}|G}{|B|} \\ &= |\mathcal{J}|G + \sum_B \omega_i - |\mathcal{J}|G = \sum_B \omega_i\end{aligned}$$

then $f \notin \mathcal{C}$.



Let $u = (1, \dots, 1)'$.

Claim. $\text{co}(\Phi) \cap \{v \in \mathbb{R}^L : v \ll -\frac{M}{|\mathcal{J}|}u\} = \emptyset$

By SFT,

$$(\forall x \in \text{co}(\Phi)) (\exists g : \mathcal{J} \mapsto \mathbb{R}^L) (\exists B \subset \mathcal{J}, |B| \leq L)$$

such that

$$x = \frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J} \setminus B} g(i) + \frac{1}{|\mathcal{J}|} \sum_{i \in B} g(i)$$

$$(\forall i \in \mathcal{J} \setminus B) \ g(i) \in \varphi(i) \tag{1}$$

$$(\forall i \in B) \ g(i) \in \text{co}(\varphi(i))$$

- $x = \frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J} \setminus B} g(i) \frac{1}{|\mathcal{J}|} \sum_{i \in B} g(i).$

- Let $y = \frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J} \setminus B} g(i).$

By Equation 1, $y \in \Phi$.

$$y - x = -\frac{1}{|\mathcal{J}|} \sum_{i \in B} g(i) \leq \frac{1}{|\mathcal{J}|} \sum_{i \in B} \omega_i \leq \frac{M}{|\mathcal{J}|} u$$

Because

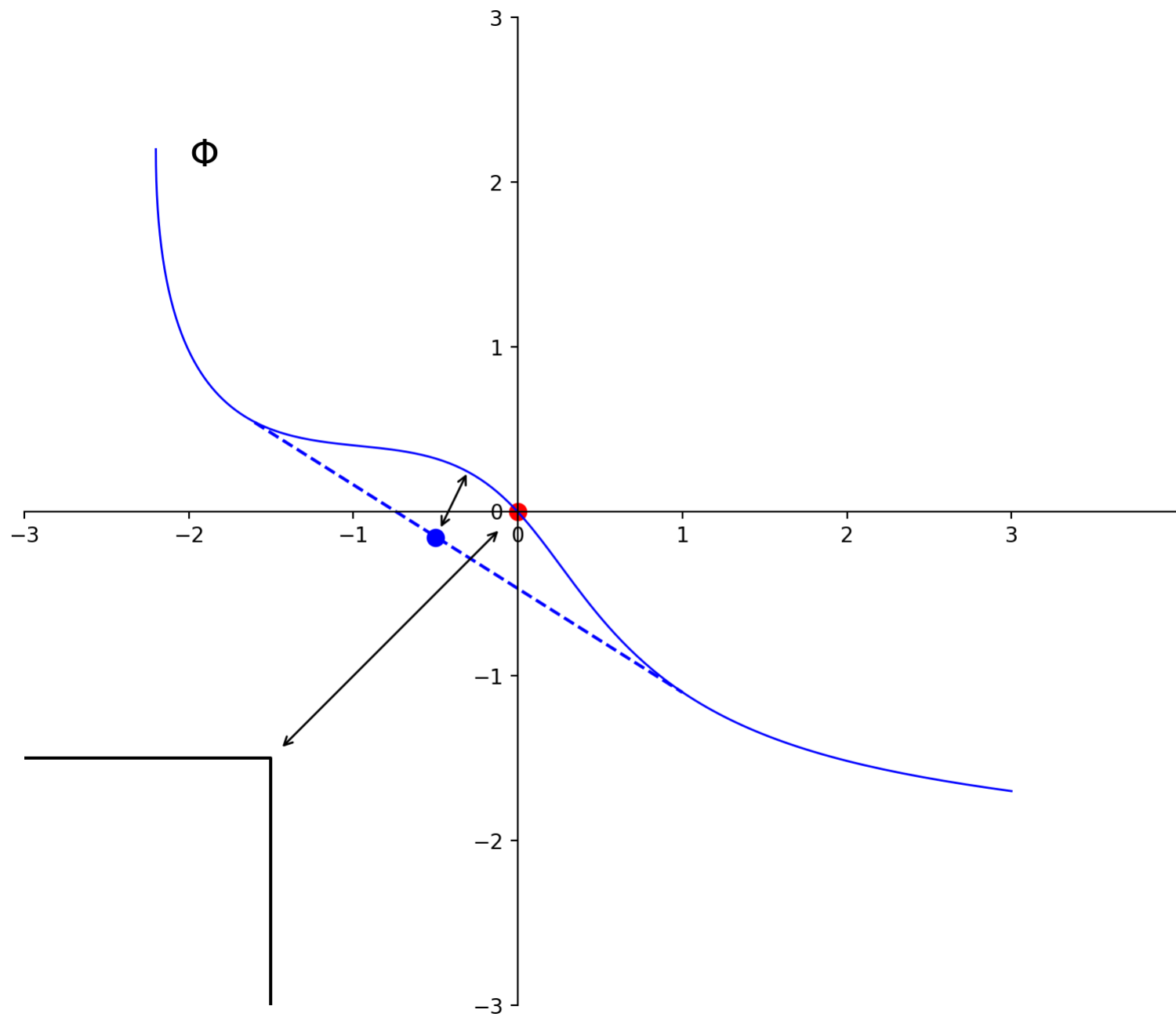
$$\varphi(i) \geq -\omega_i \implies \text{co}(\varphi(i)) \geq -\omega_i,$$

$$g(i) \in \varphi(i) \implies g(i) \geq -\omega_i \implies -g(i) \leq \omega_i;$$

and

$$\sum_{i \in B} \omega_i \leq Mu \text{ by assumption.}$$

Second claim



- By SHT and weak monotone preferences, $\exists p \in \Delta : \inf p \cdot \Phi \geq \sup\{p \cdot v : v \ll -\frac{M}{|\mathcal{J}|}u\}$
- $0 \geq \inf p \cdot \Phi \geq -\frac{M}{|\mathcal{J}|}$ because $0 \in \Phi$ and $p \cdot u = 1$.
- $(\forall m > 0) f(i) - \omega_i + \frac{u}{m} \in \varphi(i) \subset \Phi$ (by weak monotonicity).

Then

$$(\forall i \in I) \quad p \cdot [f(i) - \omega_i] \geq \inf p \cdot \varphi(i).$$

-
- Let $S = \{i \in \mathcal{J} : p \cdot [f(i) - \omega_i] < 0\}$.

$$0 > \frac{1}{|\mathcal{J}|} \sum_{i \in S} p \cdot [f(i) - \omega_i] \geq \frac{1}{|\mathcal{J}|} \sum_{i \in S} \inf p \cdot \varphi(i) \geq -\frac{M}{|\mathcal{J}|}$$

Equivalently

$$\begin{aligned} 0 &< -\frac{1}{|\mathcal{J}|} \sum_{i \in S} p \cdot [f(i) - \omega_i] \\ &\leq -\frac{1}{|\mathcal{J}|} \sum_{i \in S} \inf p \cdot \varphi(i) \leq \frac{M}{|\mathcal{J}|} \end{aligned} \tag{2}$$

-
- Since f is an allocation, $\frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J}} p \cdot [f(i) - \omega_i] = 0$. Thus $-\sum_{i \in S} p \cdot [f(i) - \omega_i] = \sum_{i \in I \setminus S} p \cdot [f(i) - \omega_i]$.

Then

$$\begin{aligned} \frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J}} |p \cdot [f(i) - \omega_i]| &= \frac{2}{|\mathcal{J}|} \sum_{i \in S} |p \cdot [f(i) - \omega_i]| \\ &\leq \frac{2M}{|\mathcal{J}|} \end{aligned}$$

- This completes the proof of 1.

- Since $0 \in \varphi(i)$, $0 \geq \frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J}} \inf p \cdot \varphi(i) = \inf p \cdot \Phi \geq -\frac{M}{|\mathcal{J}|}$.

Then

$$\begin{aligned} & \frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J}} |\inf\{p \cdot [x - \omega_i] : x \succ_i f(i)\}| \\ & \leq -\frac{1}{|\mathcal{J}|} \sum_{i \in S} [\inf p \cdot \varphi(i)] + \frac{1}{|\mathcal{J}|} \sum_{i \notin S} |p \cdot [f(i) - \omega_i]| \leq \frac{2M}{|\mathcal{J}|} \end{aligned}$$

Since $(\forall i \in S) p \cdot [f(i) - \omega_i] < 0$, $\inf\{p \cdot [x - \omega_i] : x \succ_i f(i)\} < 0$.

Thus, first term may be written in absolute value. Apply [Equation 2](#).

QED

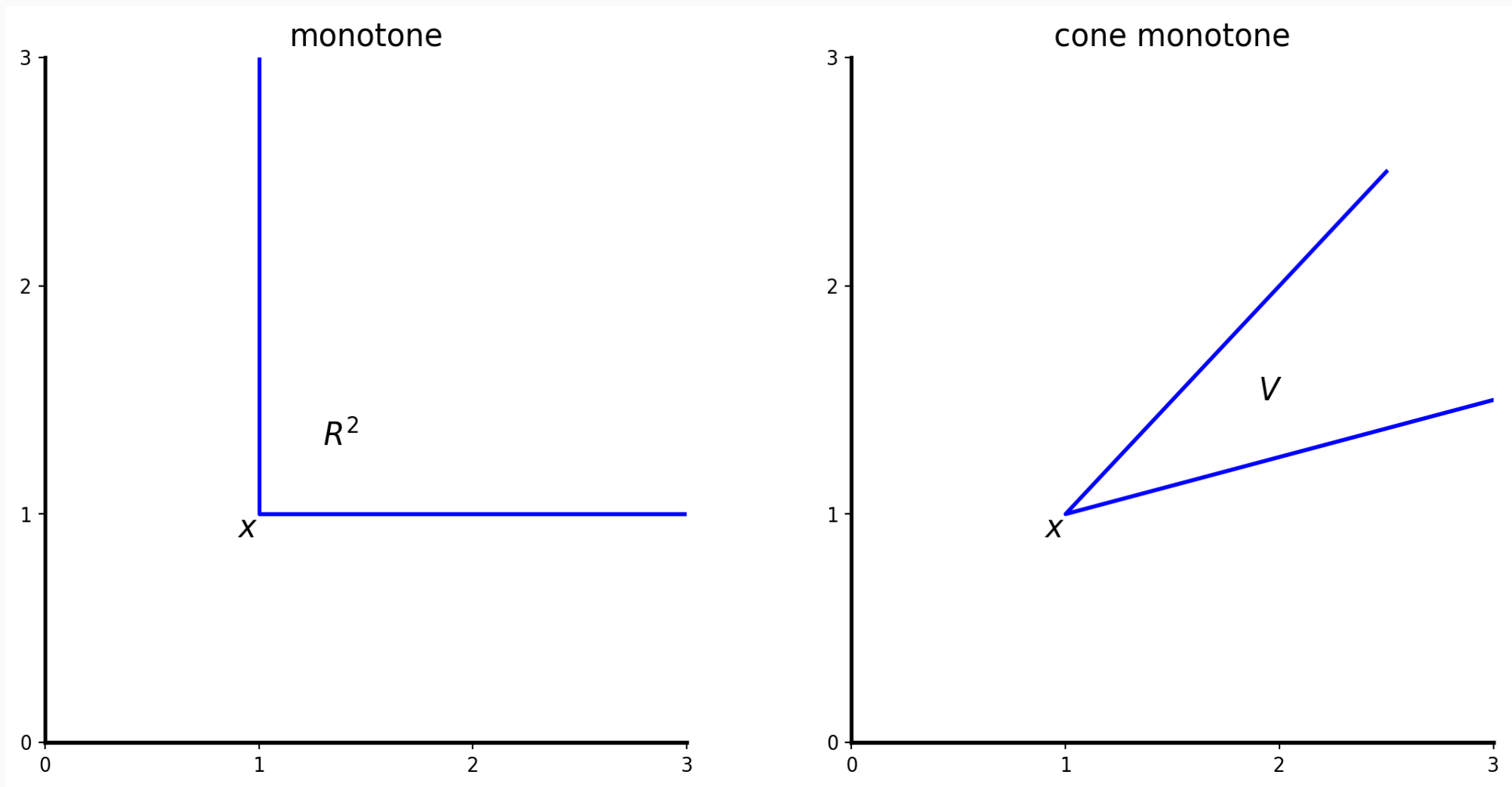
Distinct traders because of preferences

- 2 examples:
 1. Convex but non-monotone preferences.
 2. Non-convex, non-monotone preferences but uniformly bounded core allocations with respect to the size of the economy.
- The examples are in (Manelli 1991b). See also (Manelli 1991a).

Non-monotone preferences

- Preferences are cone monotone (sometimes called “proper”).
- Let $V = \{v \in \mathbb{R}^2 : \alpha(m, 1) + \beta(1, m), \alpha, \beta \in \mathbb{R}_{++}, m > 1\}$.
- In a first reading, think of $m = 2$.

Example, non-monotone preferences



1. Convex, cone-monotone preferences

- Sequence of economies with 3 “groups” of agents and 2 goods.

$$\mathcal{E}^n = \begin{cases} \text{agent 1, } \omega_1 = (0, 0), \succsim_1^n \\ \text{agent 2, } \omega_2 = (0, 0), \succsim_2^n \\ n \text{ agents type 3, } \omega_3 = (1, 1), \succsim_3 \end{cases}$$

- Agents 1’s and 2’s preferences change with the economy, their endowments do not.
- The n agents of type 3 always have the same endowment and preferences.
- $\forall n > 1, M = \sup \{ \|\omega(3) + \omega(3)\|_\infty = 2 \}$
- $\frac{M}{n} \rightarrow 0$ as $n \rightarrow \infty$.

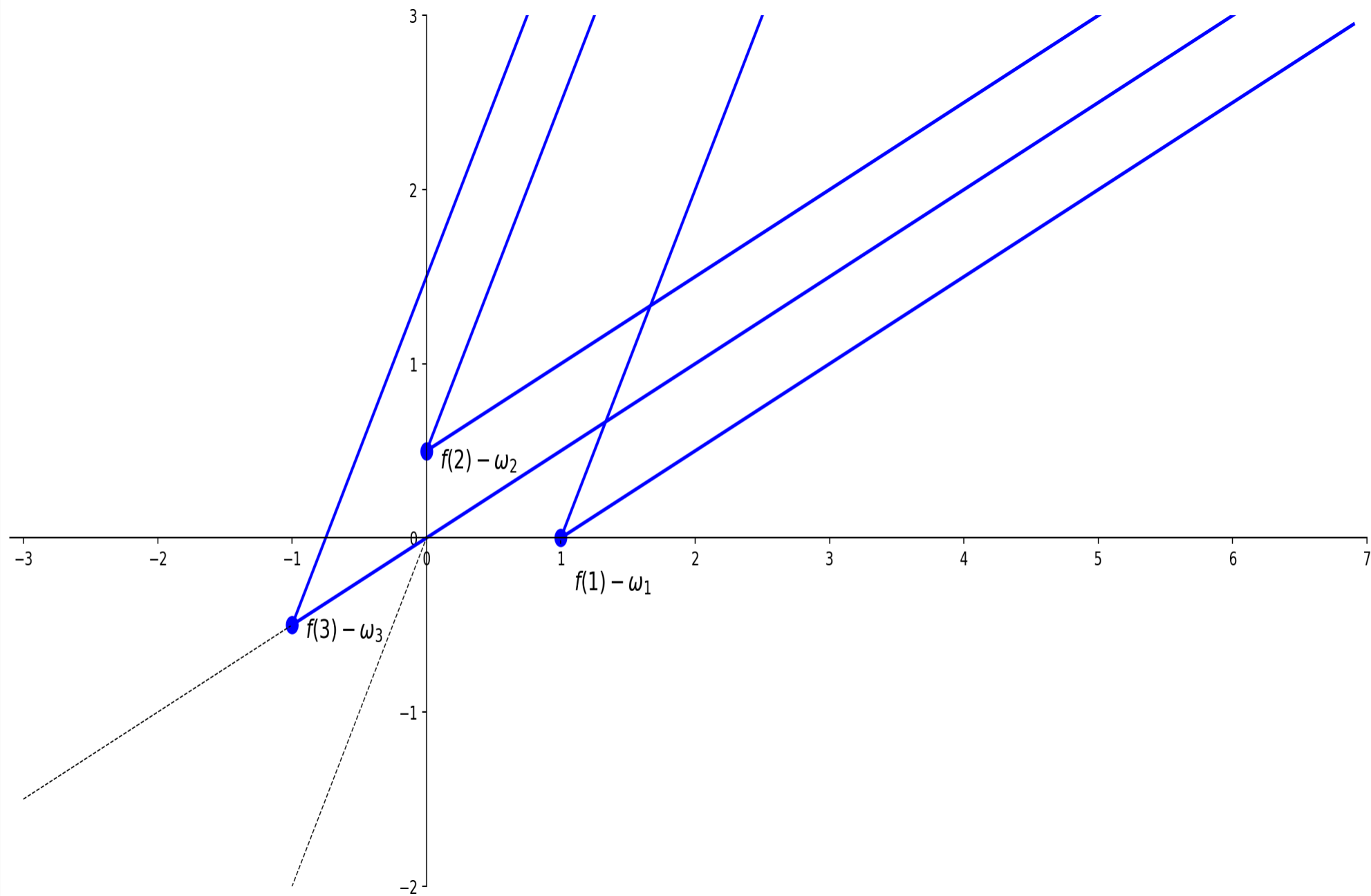
Core allocation in example 1

- Let

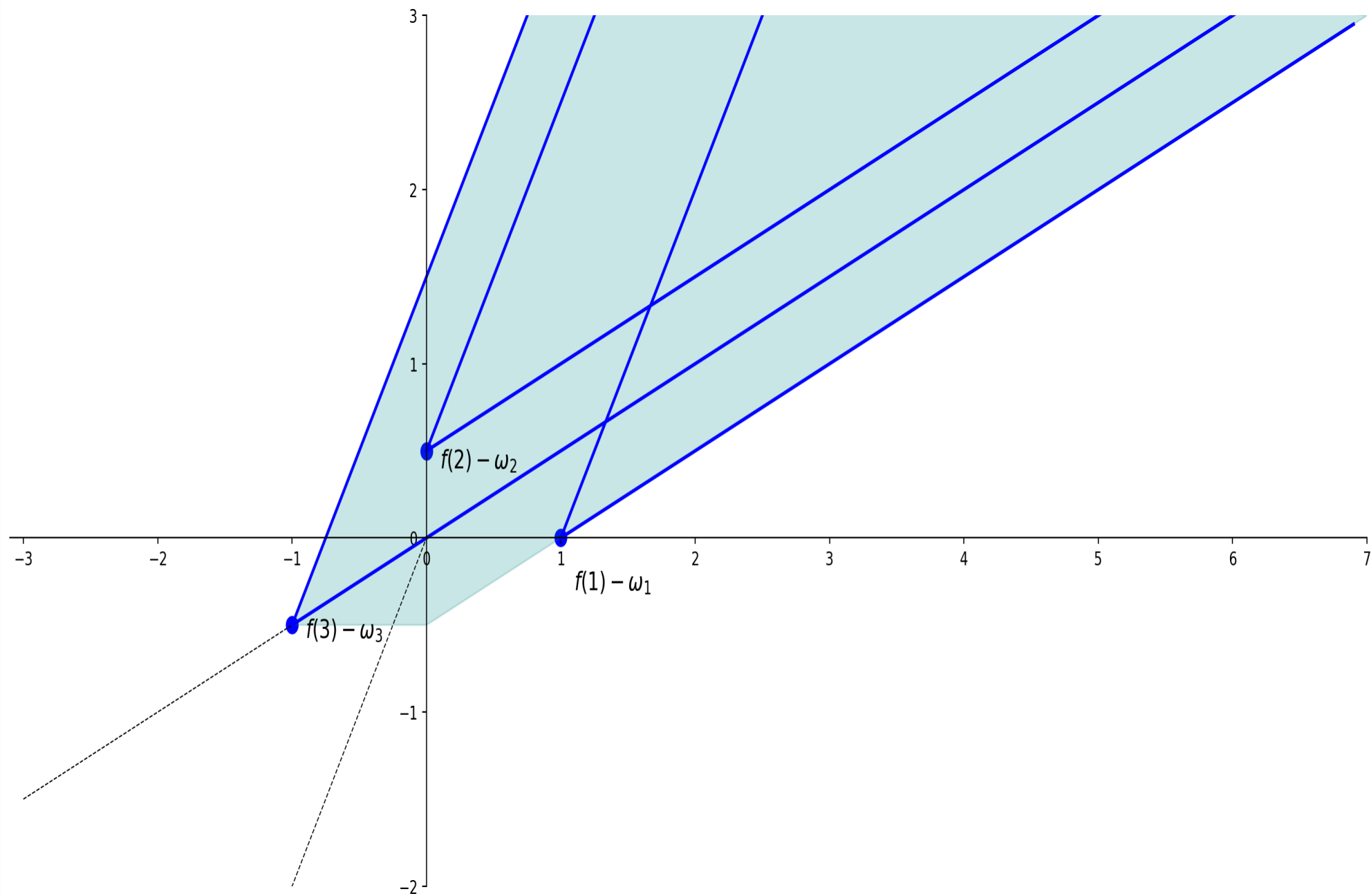
$$f^n(i) = \begin{cases} (n, 0), & \text{if } i = 1 \\ (0, \frac{n}{m}), & \text{if } i = 2 \\ (0, \frac{m-1}{m}), & \text{if } i \text{ is in group 3} \end{cases}$$

- Assume preferences satisfy $\{x : x \succ_i f^n(i)\} = f^n(i) + V$.
- Recall $\varphi(i) = \{f^n(i) - \omega_i + V\} \cup \{0\}$.
- f^n is in the core.

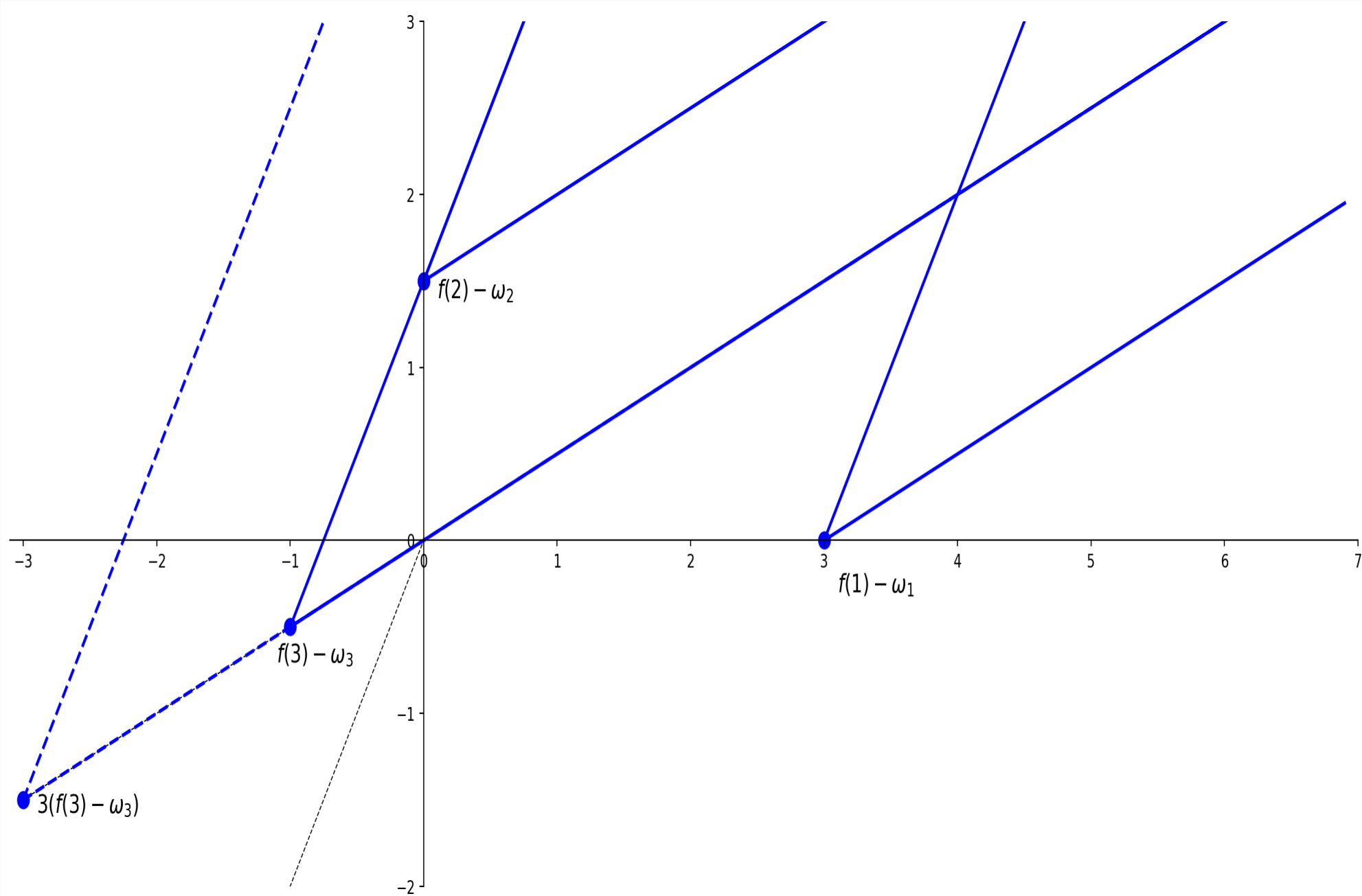
$$f^n(i), n = 1, m = 2$$



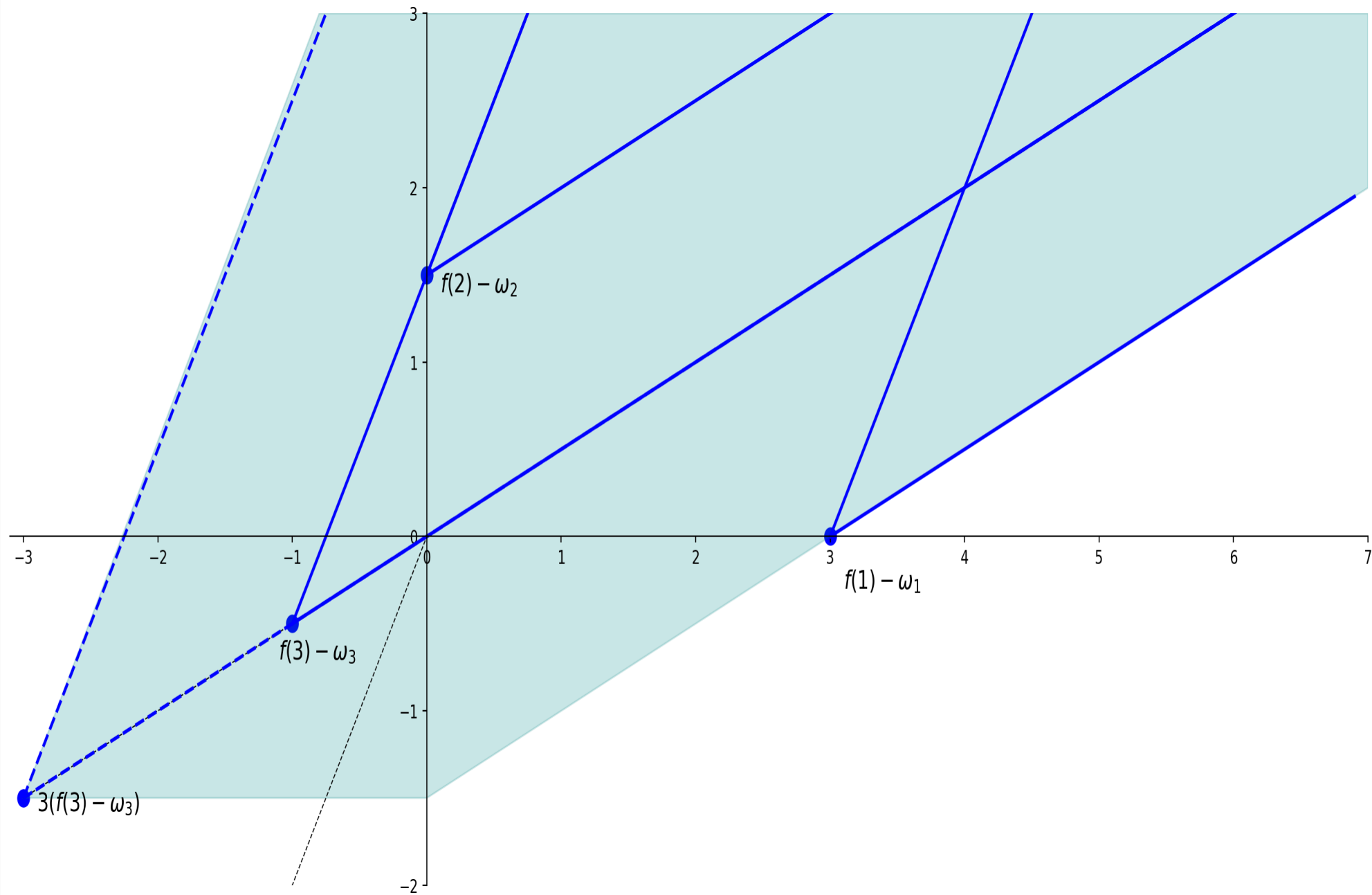
$$f^n(i), n = 1, m = 2, \text{co}(\Phi)/n$$



$$f^n(i), n = 3, m = 2$$



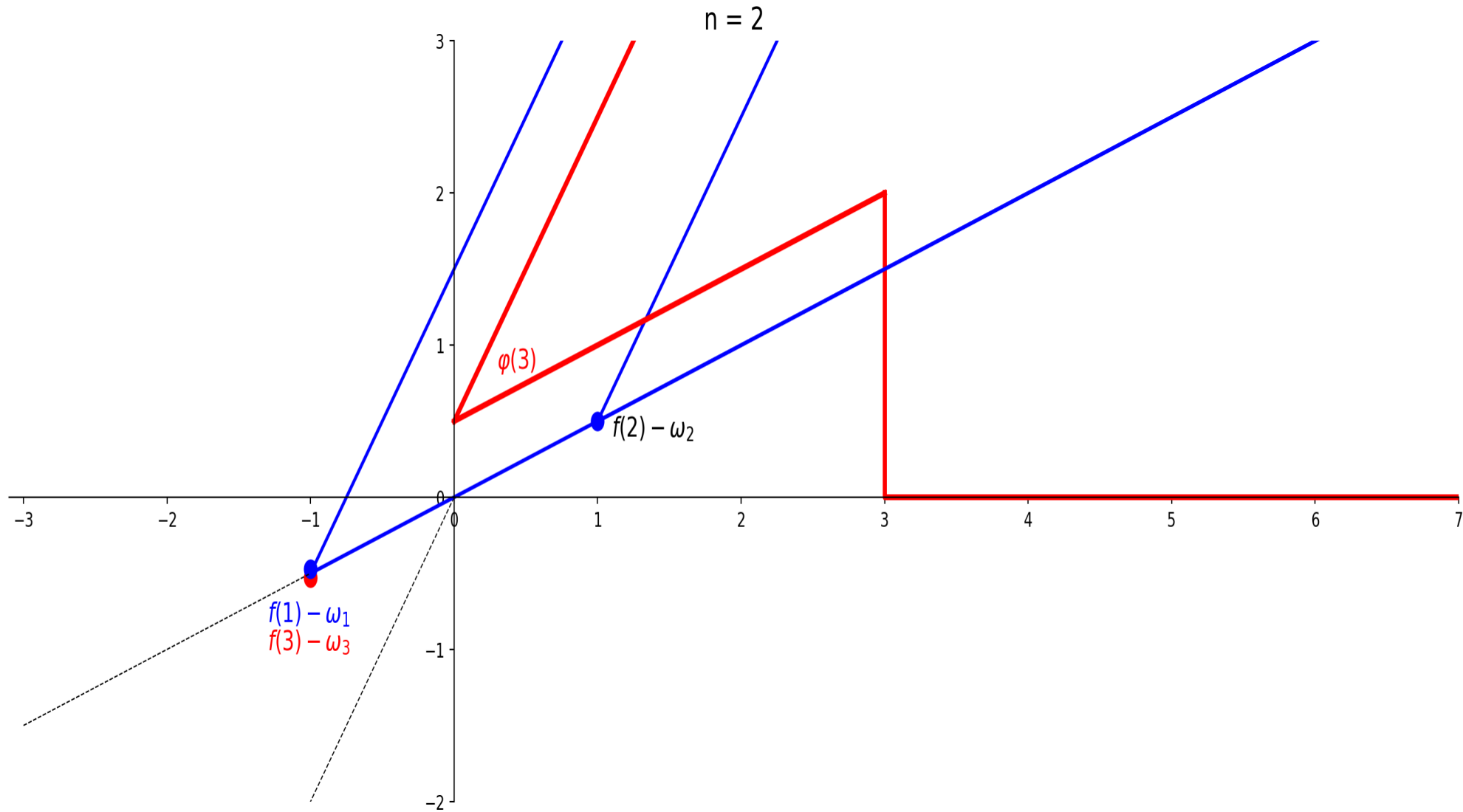
$$f^n(i), n = 3, m = 2, \text{co}(\Phi)$$



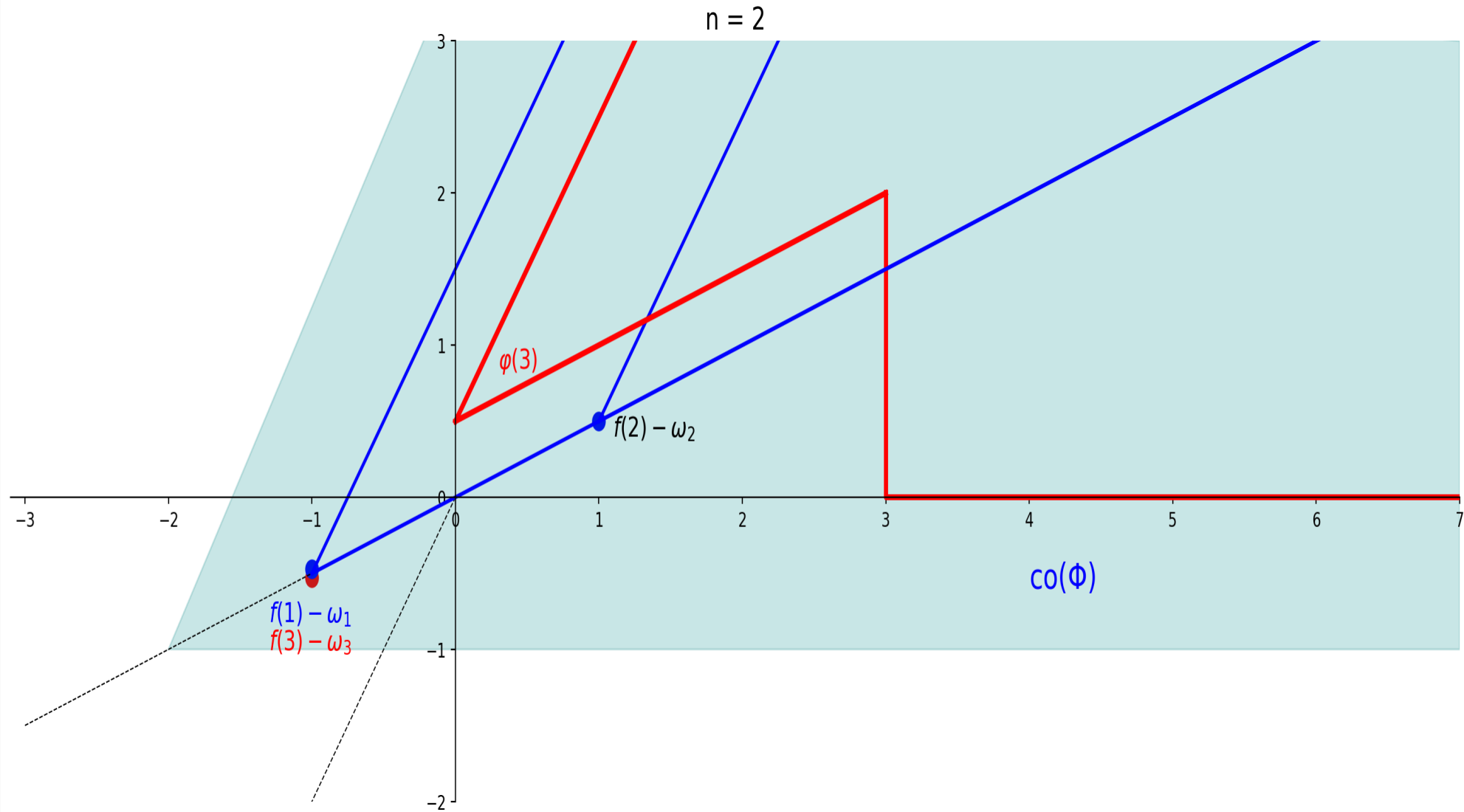
2. Non-convex, cone-monotone preferences

- The n^{th} economy has $n - 1$ agents in group 1, n agents in group 2, and a single agent in group 3.
- All agents in all economies have endowment $\omega = (1, 1)$.
- The core allocations are $f(1) = (0, 0.5)$, $f(2) = (2, 1.5)$, and $f(3) = (0, 0.5)$. *Core allocations are bounded independently of n .*
- Preferences satisfy the following conditions,
 - $\{x : x \succ_1 (0, 0.5)\} = (0, 0.5) + V$
 - $\{x : x \succ_2 (1, 1.5)\} = (2, 1.5) + V$
 - $\{x : x \succ_3 (0, 0.5)\} = \{(0, 0.5) + V\} \cup \{(n + 1, 1) + \mathbb{R}_{++}^2\}$

Non-convex, non-monotone, bounded core



Convex hull, bounded core



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